

A Mini Project Report on
EDGE DETECTION ON MEDICAL IMAGES (X-ray / MRI)

Subject Code : EE527

Submitted by

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(2025EEM1004)

Under the guidance of

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in partial fulfillment of the requirements

for the award of the degree of

MASTER OF TECHNOLOGY



DEPARTMENT OF ELECTRICAL ENGINEERING

**INDIAN INSTITUTE OF TECHNOLOGY, ROPAR
PUNJAB, 140001**

(2025-2026)

DECLARATION

I hereby proclaim that the work that is being presented in this report entitled “**Edge Detection on Medical Images (X-ray / MRI)**” is my own work and has not been reported by anyone else or submitted in any form for another degree or diploma to any other institution. Information derived from the other sources has been accredited in the text and a list of references has been given.

Prashant Kumar Gautam

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CERTIFICATE

This is to verify that the project entitled “**Edge Detection on Medical Images (X-ray / MRI)**” submitted by Prashant Kumar Gautam, for the award of the degree of M.Tech is carried out by him under my guidance.

Dr. S.K. VIPPARTHI

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VERIFICATION

This is to verify that the project entitled “**Edge Detection on Medical Images (X-ray / MRI)**” was submitted to the department of Electrical Engineering department of Indian Institute of Technology, Ropar

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ABSTRACT

Edge detection plays a critical role in medical image analysis, enabling accurate visualization of anatomical structures such as bones, tissues, and organ boundaries. This project focuses on applying advanced edge detection techniques—specifically the Canny and Laplacian operators—to medical images including X-rays and MRI scans. The Canny detector provides robust, noise-resistant edge maps through its multi-stage process, while the Laplacian operator enhances fine details by highlighting rapid intensity changes. To improve the quality of the extracted contours, morphological post-processing operations such as dilation, erosion, opening, and closing are employed to remove noise, fill gaps, and produce smoother, medically interpretable outlines. The resulting edge-enhanced images offer clearer structural boundaries that support tasks such as diagnosis, segmentation, and anatomical study. Overall, the combination of edge detection and morphological refinement demonstrates an effective approach for extracting clean and reliable contours from complex medical images.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my Teaching Assistant and guide, **Dr. S. K. Vipparthi**, Department of Electrical Engineering, IIT Ropar, for his valuable guidance, encouragement, and support throughout the duration of this mini project. His insights and constant motivation have been instrumental in the successful completion of this work. I feel thankful and express my kind gratitude towards our Director, Head of Department and all faculty members. I would also like to express my special gratitude and thanks to our parents for giving us constant support that improved my performance significantly.

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CHAPTER 1

INTRODUCTION AND PROJECT OVERVIEW

1.1 Objective

The objective of this project is to develop a simple yet effective edge-detection system capable of outlining important anatomical structures, such as bones and organs, in medical images like X-rays and MRI scans. The project aims to extract meaningful boundaries by applying classical image processing techniques instead of complex machine learning methods. Using the Canny and Laplacian edge detectors, the system identifies regions with strong intensity changes that correspond to anatomical edges. To further improve the clarity and smoothness of these edges, morphological operations are applied as a post-processing step. The goal is to generate clean contour maps that can assist in visual interpretation, segmentation preprocessing, and medical image enhancement. Overall, this project seeks to demonstrate how traditional OpenCV techniques can be used to analyze medical images efficiently.

1.2 Introduction

Medical images play a critical role in clinical diagnosis and analysis. Detecting edges helps highlight important anatomical structures such as bones, joints, and soft tissue boundaries. Edge detection is also a fundamental preprocessing step for segmentation, region extraction, and automated diagnostic systems.

This project focuses on implementing a simple yet effective edge detection pipeline using OpenCV in Python. By combining Canny and Laplacian methods and applying morphological post-processing, we aim to produce clean, continuous contours suitable for visualization and further analysis.

CHAPTER 2

METHODOLOGY

This project uses classical OpenCV image processing techniques in the following step-wise workflow:

2.1. Image Loading and Grayscale Conversion

The user inputs the file path of an X-ray or MRI image. The image is loaded directly in grayscale using `cv2.imread()` with the `IMREAD_GRAYSCALE` flag. Grayscale simplifies processing by focusing on pixel intensity values, which are essential for edge detection. Working in a single channel also reduces computational complexity.

2.2. Canny Edge Detection

Canny is a multi-stage algorithm widely used for detecting strong and weak edges with high accuracy. The process includes:

- Noise reduction using Gaussian smoothing
- Gradient computation
- Non-maximum suppression
- Hysteresis thresholding

Canny is applied with thresholds 80 (lower) and 150 (upper). These thresholds help determine which gradients form edges. The result is a binary image containing thin, continuous edge lines that typically represent bone outlines and sharp structural boundaries.

2.3. Laplacian Edge Detection

The Laplacian operator is a second-order derivative filter that detects areas where pixel intensity changes rapidly. It responds strongly to fine details and sudden intensity drops. The output image may contain negative values, so `cv2.convertScaleAbs()` is applied to convert the data into an 8-bit format suitable for visualization. Laplacian output often highlights edges that Canny may miss, especially in soft tissue or low-contrast areas.

2.4. Morphological Closing Operation

Morphological operations modify the structure of binary images based on a kernel. In this project, morphological closing is used:

- Closing = Dilation followed by erosion

A 3×3 kernel of ones is created using NumPy. Closing helps:

- Fill small holes
- Connect broken edges
- Smooth noisy contours
- Reduce small artifacts from Canny output

This produces cleaner and more continuous edges suitable for visualization.

2.5. Visualization Using Matplotlib

All results are displayed using matplotlib.pyplot in a 2×2 grid:

- Original image
- Canny edge map
- Laplacian edge map
- Morphologically cleaned edges

Matplotlib is preferred for medical imaging because it produces clear, high-quality visualizations and handles grayscale output effectively.

CHAPTER 3

IMPLEMENTATION

3.1. Implementation:

- **Image Loading:** The medical image is loaded in grayscale using `cv2.imread(..., cv2.IMREAD_GRAYSCALE)` to simplify processing and focus on intensity variations.
- **Canny Edge Detection:** Applied using `cv2.Canny(img, 80, 150)`, which computes gradient-based edges and outputs a binary edge map.
- **Laplacian Edge Detection:** Performed using `cv2.Laplacian(img, cv2.CV_64F)` to detect regions of rapid intensity change. The result is converted to 8-bit format using `cv2.convertScaleAbs()`.
- **Morphological Processing:** A 3×3 kernel (`np.ones((3,3), np.uint8)`) is used in `cv2.morphologyEx(..., cv2.MORPH_CLOSE)` to connect broken edges and clean the Canny output.
- **Visualization:** OpenCV outputs (grayscale and edge maps) are displayed using Matplotlib subplots to compare the original image, Canny edges, Laplacian edges, and the cleaned contours.

3.2. Python Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Input image from user
img_path = input("Enter the path of the medical image (X-ray/MRI): ")

# Read as grayscale
img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)

if img is None:
    print("Error: Could not load image. Check the file path!")
    exit()

# 2. Apply Canny Edge Detection
canny_edges = cv2.Canny(img, 80, 150)

# 3. Apply Laplacian Edge Detection
laplacian = cv2.Laplacian(img, cv2.CV_64F)
laplacian_edges = cv2.convertScaleAbs(laplacian)

# 4. Morphological Post-Processing
kernel = np.ones((3, 3), np.uint8)
clean_edges = cv2.morphologyEx(canny_edges, cv2.MORPH_CLOSE, kernel)

# 5. Display Using Matplotlib (Cleaner for medical images)
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.title("Original Image")
plt.imshow(img, cmap='gray')
plt.axis('off')
```

```
plt.subplot(2, 2, 2)
plt.title("Canny Edges")
plt.imshow(canny_edges, cmap='gray')
plt.axis('off')
```

```
plt.subplot(2, 2, 3)
plt.title("Laplacian Edges")
plt.imshow(laplacian_edges, cmap='gray')
plt.axis('off')
```

```
plt.subplot(2, 2, 4)
plt.title("Clean Contours (Morphological Closing)")
plt.imshow(clean_edges, cmap='gray')
plt.axis('off')
```

```
plt.tight_layout()
plt.show()
```

CHAPTER 4

RESULTS

4.1. Outputs:

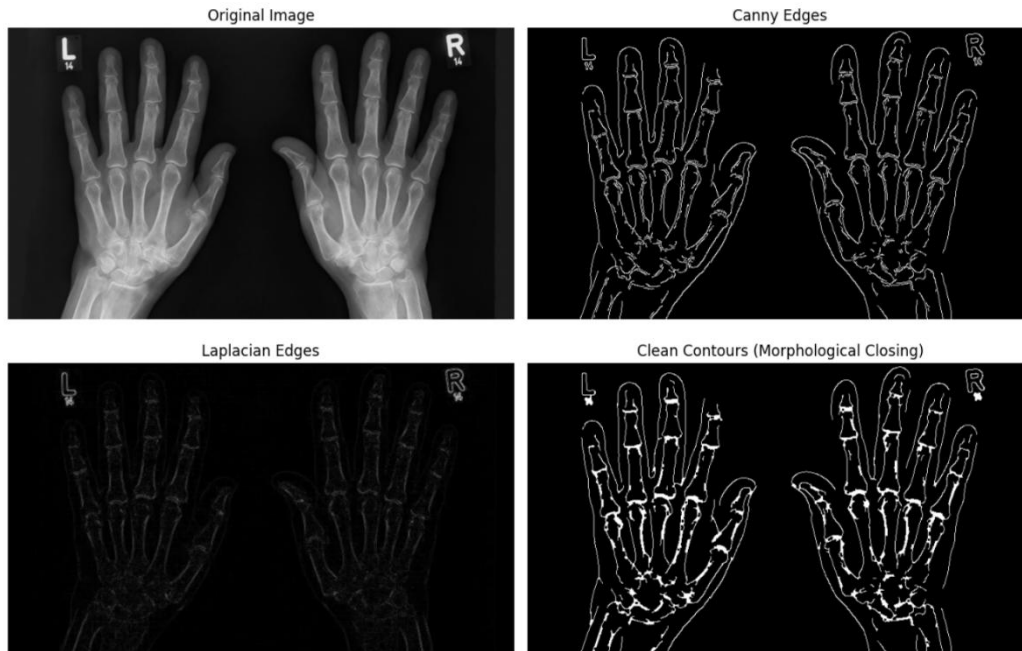


Fig. 4.1. Hand X-ray image

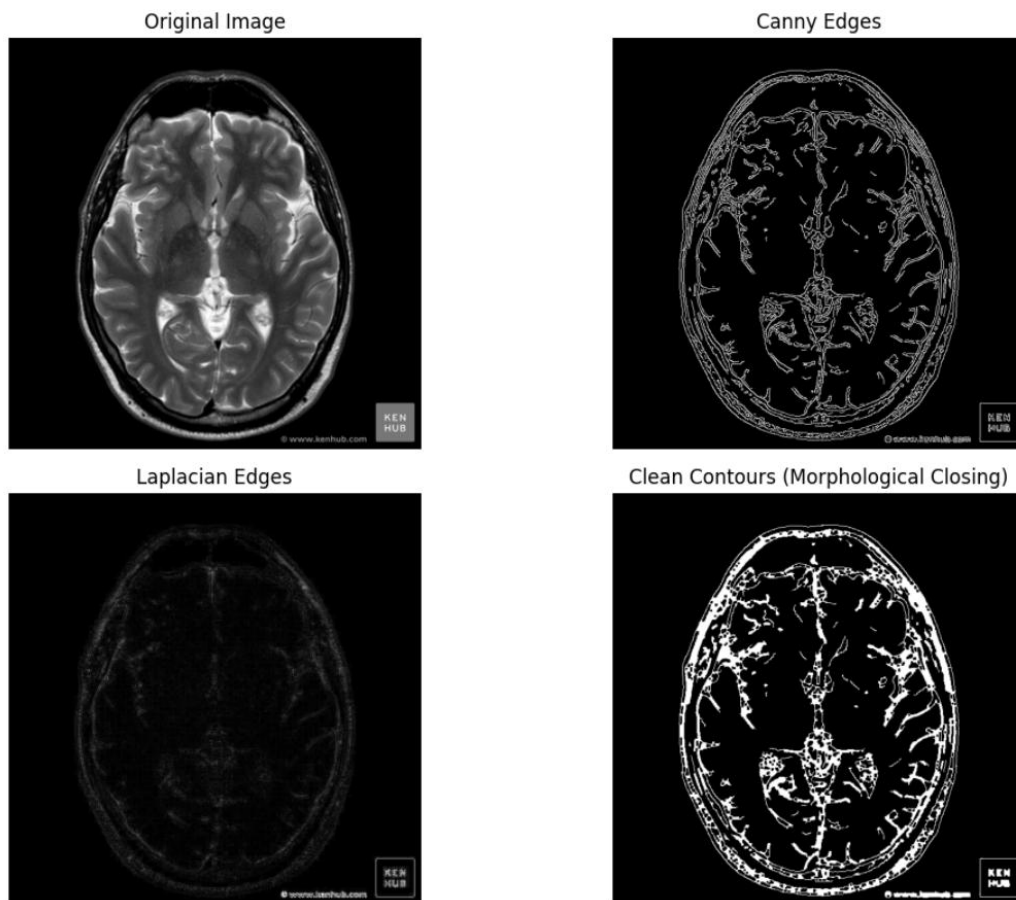


Fig. 4.2. Head MRI image

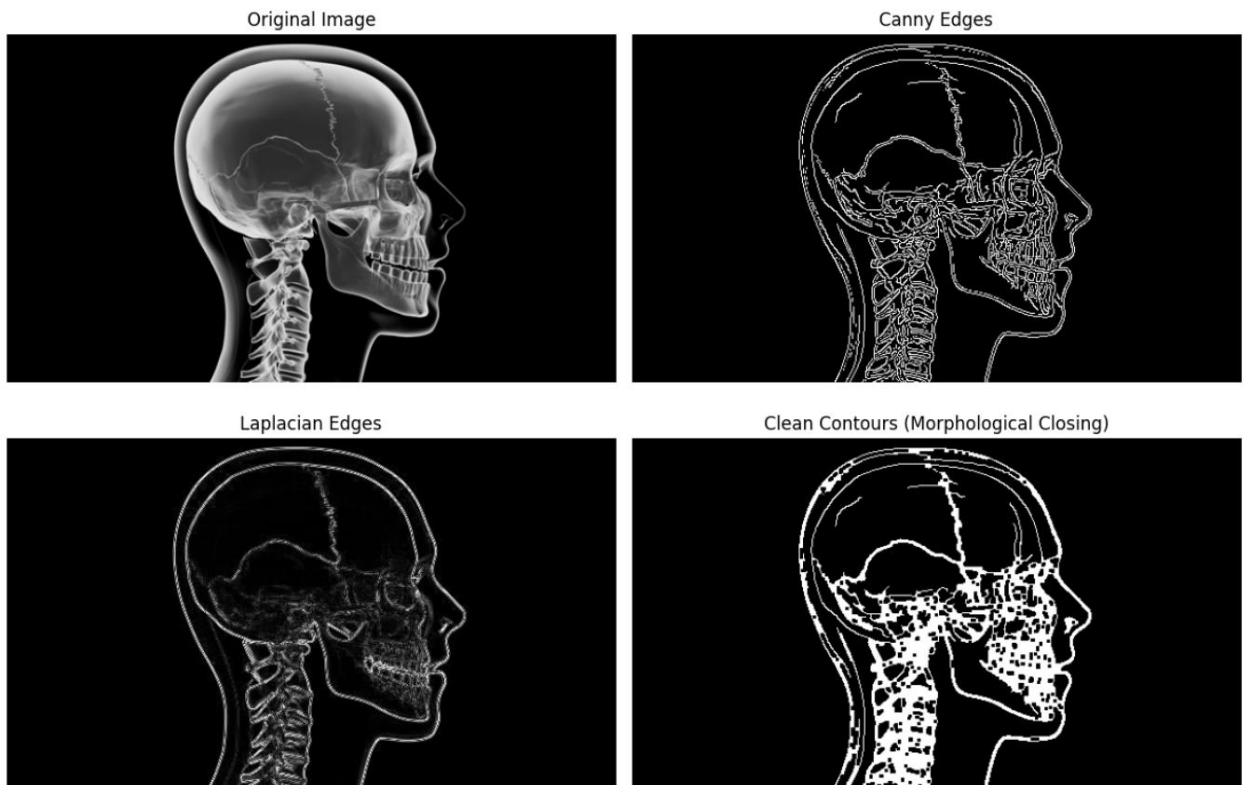


Fig. 4.3. Skull-ray image

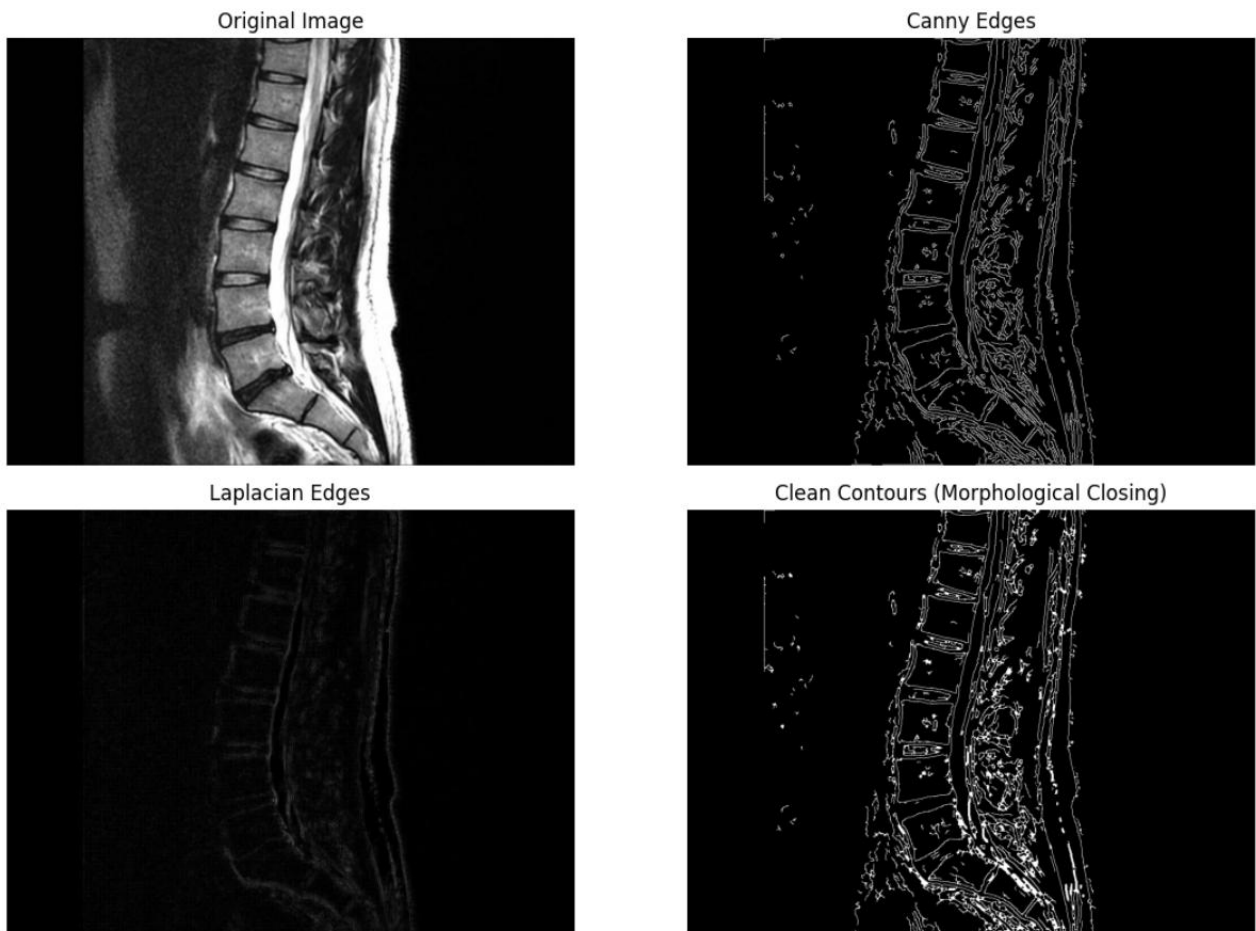


Fig. 4.4. Back-MRI image

4.2. Outcomes

- The Canny edge detector successfully highlighted strong anatomical boundaries such as bone outlines and sharp structural edges.
- Laplacian detection revealed additional subtle edges, especially in areas where intensity changes were gradual.
- Morphological closing helped connect broken edges and produced smoother, continuous contours.
- The final 2×2 Matplotlib display allowed easy visual comparison of how each processing step transformed the image.

4.3. Challenges Faced

- Some images, especially low-contrast MRIs, produced weak or faint edges.
- Noise in certain X-ray regions created small unwanted edges that required post-processing.
- Choosing appropriate Canny thresholds (80, 150) was image-dependent and required manual testing.
- Laplacian edges were sometimes thicker or noisier due to sensitivity to intensity variations.

4.4. Possible Improvements

- Apply **Gaussian blur** before edge detection to reduce noise and improve Canny results.
- Use **adaptive or dynamic thresholding** for Canny to avoid manual threshold tuning.
- Combine both edge maps (Canny + Laplacian) using `cv2.bitwise_or` for stronger overall edge detection.
- Add **morphological opening** to remove tiny artifacts or unwanted spots.
- Incorporate `cv2.findContours` and overlay detected contours on the original image for clearer visualization.
- Use **contrast enhancement** (CLAHE) to improve edges in low-contrast MRI scans.

CHAPTER 5

CONCLUSION

Through this project, I learned how classical image processing techniques can be effectively applied to medical images using OpenCV. By working with Canny and Laplacian edge detectors, I gained an understanding of how different algorithms highlight various types of edges and intensity transitions in X-ray and MRI scans. I also learned how morphological operations can refine and clean edge maps to produce smoother and more continuous contours.

Additionally, this project helped me understand the importance of parameter selection, noise handling, and visualization when analyzing medical images. Overall, the project strengthened my practical skills in Python, OpenCV, and medical image processing, and demonstrated how simple techniques can provide meaningful insights into anatomical structures.

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